MATH 152 **EXAM III** NAME _____

Mrs. Bonny Tighe PART ONE

11.8-11.11,10.5 TA_____ Wed. 12/1/2021

50 points

1. (3 points each) **Circle the letter for the correct answer**

a) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n}}{9^n (2n)!}$

A) $\pi/3$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/2$

D) $1/2$

E) Some other answer

b) The radius of convergence for the series $\sum_{n=1}^{\infty} \frac{4^n x^n}{7^{n+2}}$

A) $7/4$

B) $\{0\}$

C) $4/7$

D) $(-\infty, \infty)$

E) Some other answer

c) The interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(x+1)^n}{4^{n+1}}$

A) $[-5,3]$

B) $(-5,3)$

C) $[-5,3)$

D) $(-5,3]$

d) The power series for $f(x) = \frac{1}{x+5}$ is:

A) $\sum_{n=0}^{\infty} \frac{x^n}{5^{n+1}}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n 5^n}{x^{n+1}}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{5^{n+1}}$

D) $\sum_{n=0}^{\infty} \frac{5^n}{x^{n+1}}$

E) Some other answer

e) The third degree polynomial for the series $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{e^n n!}$ is

A) $x + \frac{x^3}{e} - \frac{x^5}{2e^2}$

B) $\frac{-x^3}{e}$

C) $x - \frac{x^3}{e}$

D) $\frac{-x^3}{e} + \frac{x^5}{2e^2} - \frac{x^7}{6e^3}$

E) Some other answer

f) The foci for the hyperbola $\frac{(x-1)^2}{7} - \frac{(y+2)^2}{9} = 1$ are

A) $(5,-2), (-3,-2)$

B) $(1 \pm \sqrt{2}, -2)$

C) $(1,2), (1,-6)$

D) $(1, -2 \pm \sqrt{2})$

E) Some other answer

g) The integral of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) C + $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) C + $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) C + $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

D) C + $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$

E) Some other answer

h) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n+1) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+2}}{(2n+2)(2n)!}$

E) Some other answer

i) Which method can be used to express $f(x) = \frac{1}{4+x}$ centered at $a = 0$ as a series?

A) Binomial Series

B) MacLaurin Series

C) Geometric Series

D) Taylor Series

E) All of the given methods can be used.

j) Which of the following is not a conic section?

A) $\frac{x^2}{4} + \frac{y^2}{7} = 1$

B) $3x^2 = 4y^2$

C) $x^2 = y$

D) $x^2 + y^2 = 1$

E) They are all conic sections.

2. (2 points each) True or False **Circle the correct answer**

a) **TRUE** **FALSE** The directrix for the parabola $x^2 - 4x - 4y =$ is $y = -2$

b) **TRUE** **FALSE** When using the Ratio Test on a power series, the limit $L = 0$ means that the Radius of convergence is 0.

c) **TRUE** **FALSE** The asymptotes for the hyperbola $\frac{(x-1)^2}{16} - \frac{(y+2)^2}{25} = 1$ are $y = \pm\frac{5}{4}x$

d) **TRUE** **FALSE** The Maclaurin series representation for $f(x) = x\cos(x^3)$ is $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+1}}{(2n)!}$.

e) **TRUE** **FALSE** The foci of the hyperbola $4x^2 - 4(y-1) = 36$ are $(\pm 3\sqrt{2}, 1)$

f) **TRUE** **FALSE** The Taylor series for $f(x) = \sin x$ at $x = \pi/6$ is $\sum_{n=0}^{\infty} \frac{(-1)^n (x - \pi/6)^{2n+1}}{(2n+1)!}$.

g) **TRUE** **FALSE** $e = 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \ldots$

h) **TRUE** **FALSE** A parabola can never intersect its diretrix.

i) **TRUE** **FALSE** To find the foci of a hyperbola, use the formula $c^2 + b^2 = a^2$

j) **TRUE** **FALSE** The radius of convergence for the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{4^n n!}$ is 4.

3. (1 point each) True of False **Circle the correct answer**

a) **TRUE FALSE** $9x^2 + 4x = y^2 + 6$ is an ellipse.

b) **TRUE FALSE** $\frac{9!}{9} = 8!$

c) **TRUE FALSE** We can use the geometric series test to find the radius and interval of convergence for $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{5^{n+1}}$

d) **TRUE FALSE** The endpoints of the Interval of Convergence for a series must always be tested to see if they are included or not.

e) **TRUE FALSE** $T_3(x)$ is the third partial sum for a Taylor Series.

MATH 152 **EXAM III** NAME _____

Mrs. Bonny Tighe PART TWO

11.8-11.1,10.5 TA_____ Wed.12/1/2021

50 points

1. (5 points) Find the vertex, focus, and directrix. Sketch the graph

$x^2 + 4x + 2y + 6 = 0$

2. (7 points) Use the Binomial Theorem to find the function as a power series.

$$f(x) = \frac{1}{\sqrt[3]{1 - 4x^2}}$$

3. (5 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$$9(x+2)^2 - 4(y-1)^2 = 36$$

4. (5 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{3^n (n^2+2)}$$

5. (5 points) Find the equation of the ellipse with foci at (2,5) and (2, 3) and vertex at (2, 6).

6. (8 points) Find the Taylor series for $f(x) = \frac{1}{(2+x)^3}$ centered at

a = 2

7. (6 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \sqrt{x}$ centered at $a = 4$
.

b) (4 points) Use it to approximate $\sqrt{3.9}$.

c) (5 points) If $\left|f^{(n+1)}(x)\right| \le M$ for $|x - a| \le d$, then the remainder

$R_n(x) \le \dfrac{M}{(n+1)!}\left|x - a\right|^{n+1}$ for $|x - a| \le d$

How accurate is this approximation when $3.5 < x < 4.5$

8. (5 points) Evaluate the integral as a power series to three non-zero terms.

$$\int_0^{0.5} \frac{x}{5 - x^3}\,dx$$

MATH 152 **EXAM IIIA** NAME _____

Mrs. Bonny Tighe PART ONE

11.8-11.11, 10.5 TA_____Wed.12/1/2021

50 points

1. (3 points each) **Circle the letter for the correct answer**

a) The series which is equivalent to $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{5^n n!}$

A) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{5^n n!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{5^{n+1} (n+1)!}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^{n-1} x^{2n-1}}{5^{n-1} (n-1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^{n-1} x^{2n+2}}{5^{n+1} (n+1)!}$

E) Some other answer

b) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n}}{9^n (2n)!}$

A) $\pi/3$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/2$

D) $1/2$

E) Some other answer

c) Which of the following is not a conic section?

A) $\frac{x^2}{4} + \frac{y^2}{7} = 1$

B) $3x^2 = 4y^2$

C) $x^2 = y$

D) $x^2 + y^2 = 1$

E) They are all conic sections.

d) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n+1) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+2}}{(2n+2)(2n)!}$

E) Some other answer

e) The third degree polynomial for the series $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{e^n n!}$ is

A) $x - \frac{x^3}{e} + \frac{x^5}{e^2(2)}$

B) $1 - \frac{x}{e} + \frac{x^2}{e^2} - \frac{x^3}{e^3(6)}$

C) $\frac{-x^3}{e}$

D) $1 - \frac{x^3}{e} + \frac{x^5}{2e^2}$

E) Some other answer

f) The foci for the hyperbola $\frac{(x-1)^2}{7} - \frac{(y+2)^2}{9} = 1$ are

A) $(5,-2), (-3,-2)$

B) $(1 \pm \sqrt{2}, -2)$

C) $(1,2), (1,-6)$

D) $(1, -2 \pm \sqrt{2})$

E) Some other answer

g) The integral of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) C + $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) C + $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) C + $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

D) C + $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$

E) Some other answer

h) The radius of convergence for the series $\sum_{n=1}^{\infty} \frac{4^n x^n}{7^{n+2}}$

A) $7/4$

B) $\{0\}$

C) $4/7$

D) $(-\infty, \infty)$

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C) $[-5,3)$

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j) Which method can be used to express $f(x) = \frac{1}{4+x}$ as a series?

A) Binomial Series

B) MacLaurin Series

C) Geometric Series

D) Taylor Series

E) All of the given methods can be used.

2. (2 points each) True or False **Circle the correct answer**

a) **TRUE** **FALSE** The Taylor series for $f(x) = \sin x$ at $x = \pi/6$ is $\sum_{n=0}^{\infty} \frac{(-1)^n (x - \pi/6)^{2n+1}}{(2n+1)!}$

b) **TRUE** **FALSE** When using the Ratio Test on a power series, the limit $L = 0$ means that the Radius of convergence is 0.

c) **TRUE** **FALSE** The asymptotes for the hyperbola $\frac{(x-1)^2}{16} - \frac{(y+2)^2}{25} = 1$ are $y = \pm\frac{5}{4}x$

d) **TRUE** **FALSE** The Maclaurin series representation for $f(x) = x\cos(x^3)$ is $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+1}}{(2n)!}$.

e) **TRUE** **FALSE** The foci of the hyperbola $4x^2 - 4(y-1) = 36$ are $(\pm 3\sqrt{2}, 1)$

f). **TRUE** **FALSE** The directrix for the parabola $x^2 - 4x - 4y =$ is $y = -2$

g) **TRUE** **FALSE** $e = 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \ldots$

h) **TRUE** **FALSE** A parabola can never intersect its diretrix.

i) **TRUE** **FALSE** To find the foci of a hyperbola, use the formula $c^2 + b^2 = a^2$

j) **TRUE** **FALSE** The radius of convergence for the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{4^n n!}$ is 4.

3. (1 point each) True of False **Circle the correct answer**

a) **TRUE** **FALSE** $9x^2 + 4x = 4y^2 + 6$ is an ellipse.

b) **TRUE** **FALSE** $\frac{11!}{11} = 10!$

c) **TRUE** **FALSE** We can use the geometric series test to find the radius and interval of convergence for $\sum_{n=0}^{\infty} \frac{(-1)^n x^{3n}}{5^{n+1}}$

d) **TRUE** **FALSE** The endpoints of the Interval of Convergence for a series must always be tested to see if they are included or not.

e) **TRUE** **FALSE** $T_3(x)$ is the third partial sum for a Taylor Series.

MATH 152 **EXAM IIIA** NAME _____

Mrs. Bonny Tighe PART TWO

11.8-11.11, 10.5 TA_____ Wed.12/1/2021

50 points

1. (5 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \frac{1}{x\sqrt{x}}$ centered at $a = 4$

b) (4 points) Use it to approximate $\sqrt{4.1}$.

c) (4 points) How accurate is this approximation when $|x-9|<0.5$

If $\left|f^{(n+1)}(x)\right| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a| \le d$

2. (5 points) Find the equation of the ellipse with and focus at $(1,2)$ and vertices at $(3,2)$ and $(-3,2)$

3. (6 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{\sqrt{n+3}}$$

4. (5 points) Find the vertex, focus, and directrix. Sketch the graph

$y^2 + 4y + 4x + 8 = 0$

5. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$4(y - 2)^2 - 9(x - 1)^2 = 36$

5. (8 points) Use the Binomial Theorem to find the function as a power series.

$$f(x) = \frac{1}{\sqrt[3]{1 - 4x^2}}$$

6. (6 points) Evaluate the integral as a power series and approximate to three non zero terms. $\int_0^{0.2} \frac{x^2}{4 + x^2}\, dx$

7. (8 points) Find the Taylor series for $f(x) = \frac{1}{(x - 1)^3}$ centered at a = 3